Zehan Ruan

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SUMMARY OF QUALIFICATIONS

- Creativity developed in Building an AI visual obstacle avoidance car.
- Electronics & programming skills via self-built drone, AI car, rocket (PBL projects).
- Theoretical knowledge: AMC12 Distinction, physics/chemistry awards, C++ OI.
- Electrical circuit experience shown on Self-made FPV drone.
- Resource searching ability shown on finishing two PBL without any teachers' help and got 5 on 7 AP exam with self-study.
- Problem-solving: Overcame hardware failures in drone/rocket builds.
- Language: English, Chinese, Spanish

EXPERIENCE

Teacher, Students' parents, Wenzhou City, Zhejiang Province, China March-April 2026

- Accessed to the student's education system to provide a learning method that matched with both of this education rules and student's learning habit, so that my class could reach to the highest efficiency.

Dessert shop's manager, Ruian High School, Wenzhou City, Zhejiang Province, China October 2025

- Did a survey about which kind of dessert is the most popular one from the students
- According to the survey's result, prepared the dessert's material and made them up to a half-processed dessert in advance
- Divided jobs and assigned them to my workers

Tech Club Founder &President, Ruian High School, Wenzhou City, Zhejiang Province, China (2023-2026)

• Led Raspberry Pi, drone, rocket, Android app projects; inspired student game dev.

PBL Project Lead – Custom Drone, Ruian High School, Wenzhou City, Zhejiang Province, China (Grade 11-12)

• Designed, soldered, coded, tuned quadcopter (PID, GPS, FPV); overcame crashes/explosions.

PBL Project Lead – AI Avoidance Car & Rocket, Ruian High School, Wenzhou City, Zhejiang Province, China (2024-2025)

• Built embedded car with AI recognition; synthesized rocket fuel, Arduino data logging.

Android App Developer (Tomato Clock), Personal (2024)

• Developed cross-device to-do app with server simulation.

EDUCATION

Candidate for Bachelor of Applied Science in Electrical Engineering
September 2026-present
University of Waterloo, Waterloo, Ontario

High School Diploma June 2026
Ruian High School, Wenzhou City, Zhejiang Province, China

Relevant Courses: Physics, Calculus, Chemical, Biology, Statistic, Computer Science

RELEVANT PROJECTS

Custom FPV Drone, PBL (2024-2025)
• Independently assembled high-speed long-range quadcopter; PID tuning, wireless, night flight add-ons.

AI Avoidance Smart Car, Qingdao AI PBL (2024)
• Group leader: Python control, camera AI training (100 images), obstacle avoidance.

KNSB Rocket Test Rig, PBL (2025)
• Led fuel synthesis, 3D printed body, Arduino 80Hz pressure data logging.

Android Tomato Clock App (2024)
• Full stack: Data sync server (Linux sim), to-do list, notifications.

Personal School Website (2024)
• MySQL backend, AI summary charts for student scores.

C++ Multiplayer SanGuoSha (2023)
• Complex logic, networking.

AWARDS AND CERTIFICATES

American Mathematics Competitions 12 (AMC12), Mathematical Association of America, USA 2024-2025
• Certificate of Distinction, 2nd Place, Excellence &AIME Qualifier; advanced math problem solving
Euclid Mathematics Contest, University of Waterloo, Canada Grade 11
• Global Top 25%; mathematical reasoning
JCCO Canadian Chemistry Olympiad, Canada Grade 10
• National Gold, Global Bronze; chemistry knowledge
CCC Canadian Chemistry Contest, Canada Grade 11
• Regional Excellence; advanced chemistry
Australian Mathematics Competition (Kangaroo), Australia Grade 10
• Gold Award; logical thinking
Australian Mathematics Competition (AMC), Australia Grade 10
• National First Prize; math proficiency
British Physics Olympiad, UK Grade 10
• Global Bronze; physics knowledge
ACSL American Computer Science League, USA Grade 11
• National Qualifier; CS &logic
Zhejiang C++ Informatics Olympiad, China High School
• Second Prize; programming skills
AP Computer Science A, College Board, USA High School

• Score 5; Java programming

ACTIVITIES AND INTERESTS

- ***Harvard Summit (Computer Biology &Bioengineering)**, Harvard University, USA*
  Summer 2023
  • Participant; explored brain-machine interfaces and bio-electronics
- ***Kunshan Duke University Math Modeling Contest**, Kunshan Duke University, China*
  2024
  • Competitor; solved Rubik's cube optimal steps using breadth-first search
- ***Technology Club**, Ruian High School, Wenzhou, Zhejiang, China 2023-2026*
  • Founder &President; led drone, rocket, AI car, app development projects